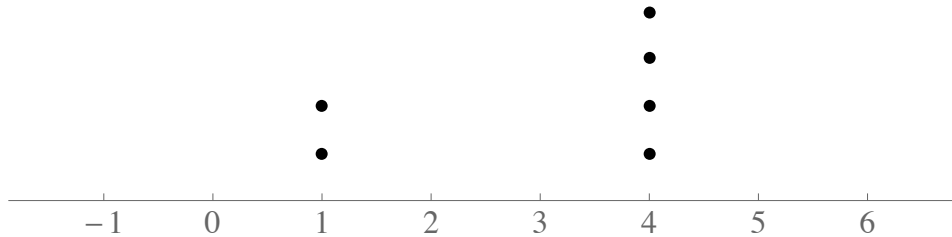


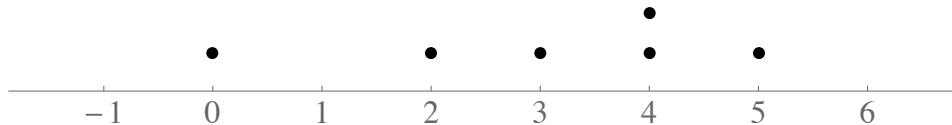
Chip-Firing Games

Fall 2018 ARML Power Contest

Consider an ordinary number line (only the integers will be of concern) where a number of counters or “chips” have been placed on some of the points. On the line below, two chips have been placed on the 1-point and four chips on the 4-point.



This configuration will be described by saying that the “1-pile has two chips, and the 4-pile has four chips.” The *chip-firing game* proceeds by sending chips from one point to another according to the following simple rule: if a pile has two or more chips in it, that pile may “fire” one chip to each of its neighbors. In the example above, each of the two piles may be fired, and the result of firing both is the arrangement below.



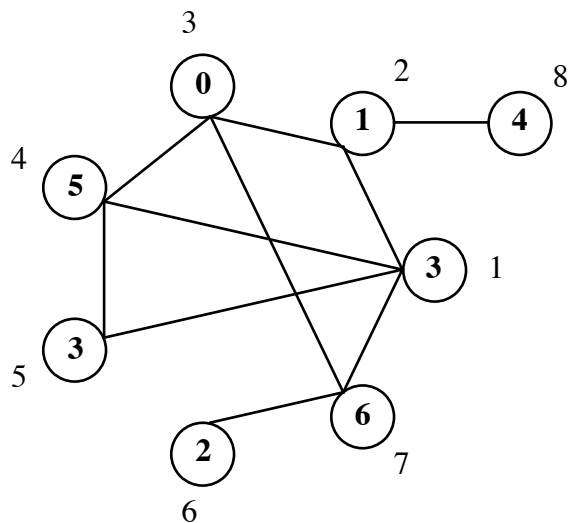
At this point, the 4-pile could be fired again, leaving it empty but putting a second chip on the 3-pile and the 5-pile, and the game could continue from there.

If all the piles in a configuration have one chip (or no chips, allowing empty piles) the configuration will be called *stable*. The game will end if it reaches a stable configuration, because there is nothing else to do. All starting configurations eventually reach a stable configuration. It turns out that the ending configuration doesn’t depend on the order in which the piles are fired. That is, if two or more piles can be fired, they can be fired in any order, including firing other piles that weren’t fireable but became so during the process, and the final stable configuration of chips will always be the same (we will prove this later). For example, in the original configuration, the game could have begun by firing the 4-pile twice, leaving two chips in each of the 1-, 3-, and 5-piles. Then the 3-pile could be fired, even before the 1-pile which was originally fireable, even though the 3-pile wasn’t. All orders of firing lead to the same final stable configuration.

- [3] **Problem 1.** Determine the final configuration resulting from the example, and the total number of firings needed to obtain it. (By total it is meant to count each and every firing, counting multiple times if a pile is fired more than once.)

- [3] **Problem 2.** Let k be a positive integer, and place one chip in each pile from the 1-pile up to the k -pile. Then place a second chip in the 1-pile. Determine the final stable configuration resulting from this starting configuration, and the total number of firings that occur. (You do not need to prove your result, just describe it. The necessary induction proof for this problem will be considered “obvious.”)
- [3] **Problem 3.** For n from 1 to 6, determine the stable configuration resulting from starting with n chips in the 0-pile and no other chips in any other pile.
- [2] **Problem 4.** Determine the number of firings that occur in each case of the previous problem.
- [4] **Problem 5.** State and prove a conjecture describing the stable configuration resulting from starting with an arbitrary number, n , of chips in the 0-pile and no other chips.
- [2] **Problem 6.** State and prove a formula (explicit or recursive) involving n that will compute the number of firings that occur in the previous problem.

Instead of just the integers on a number line, the chip-firing game can be played on a more general game board. A game board consists of a number of piles, $P_1, P_2, \dots, P_n$, and a list of connections. A connection tells when two piles should be considered “next to” each other. That is, if P_3P_7 is one of the connections, then if pile P_3 fires, one of its chips will go to pile P_7 , and vice versa. We might draw a game board as seen below. Piles are represented as circles, and the connections as lines or curves from one circle to another. The boldfaced numbers inside the circles will represent the number of chips in that pile, while the number outside the pile simply represents the pile number. Piles P_1 and P_5 each have three chips in them in the diagram, while pile P_3 has no chips at present.



The game board will be required to be connected in the sense that there is a sequence of connections from any one pile to any other. That is, it must be possible for a chip starting

in any pile to end up in any other pile if it is fired enough times from one pile to a connected pile. The game also uses only a finite number of chips.

The number of connections to any given pile is called its *valence*. In the example, P_1 has valence 4 while P_8 has valence 1. The firing rule in the game played on a general board is that any pile which currently has at least as many chips as its valence may fire one chip to each of its neighbors. Thus, P_1, P_2 , and P_3 in the diagram may not fire, while all the rest of the piles can. If P_5 fires, the result is that P_1 will now have four chips (and now be able to fire), P_4 will have six, and P_5 will drop to one, while no other pile is changed.

[2] **Problem 7.** Draw the diagram that would result from the example if every pile that can fire initially does so, while no pile that could not fire initially (such as P_1) does so, even if it becomes able to. It may be assumed that firing order does not matter.

[2] **Problem 8.** Prove that, from any configuration of chips on any game board, if it is possible to fire both piles P_x and P_y then the result of firing P_x first and then firing P_y is the same as firing P_y first then P_x . (This, along with some induction, allows you to rearrange any firing order into any other that fires the same piles, justifying the assumption in the previous problem.)

For convenience, let P be the total number of piles, C the total number of connections, and N the total number of chips in the game. In the example game board, $P = 8, C = 10$, and $N = 24$.

Problem 9.

[2] (a) Show that if the total number of chips on the game board is more than $2C - P$ than the game will never reach a stable configuration. That is, show that there must always be some pile that contains more chips than its valence.

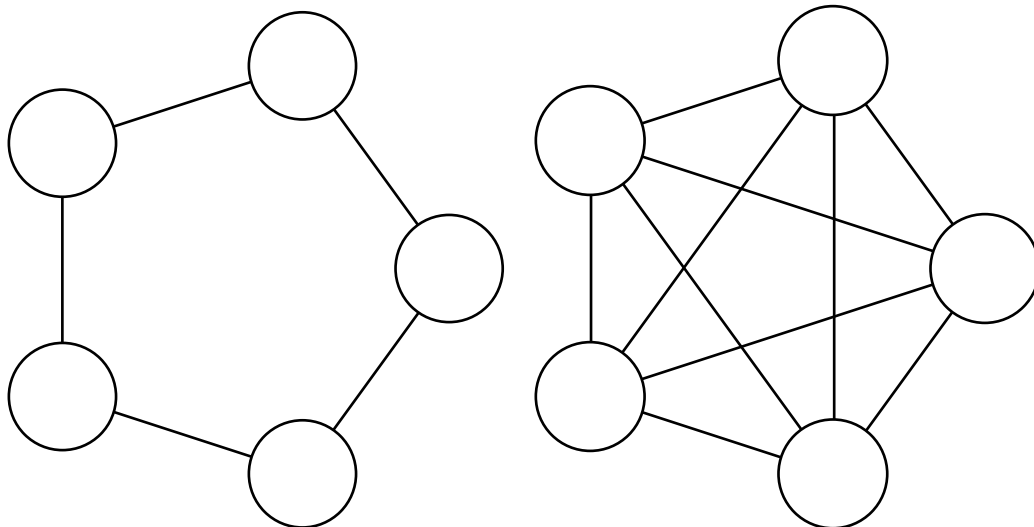
[2] (b) Show that this bound is tight. That is, show that for any game board there is an initial configuration with exactly $2C - P$ total chips for which the game is in a stable configuration.

[3] **Problem 10.** Prove that if a game never reaches a stable configuration then *every* pile must fire infinitely often.

[3] **Problem 11.** Prove that if a game does reach a stable configuration then there is at least one pile that *never* fires.

Problem 9 showed that if there are more than $2C - P$ chips on the board the game must go on forever without reaching a stable configuration. There is a similar theorem, stating that if there are fewer than C chips on the board then the game will always reach a stable configuration, and this bound is tight in that if there are at least C chips in the game then there are some configurations that never become stable. This will not be proved (but see the article referenced at the end of this contest packet).

- [2] **Problem 12.** The game board below left has $C = 5$ connections. Find, with justification, an initial configuration of five chips which never leads to a stable configuration.



- [2] **Problem 13.** The game board above right has $C = 10$ connections. Find, with justification, an initial configuration of ten chips which never leads to a stable configuration.

A game board where every pile is connected to every other pile is called *complete*. The board in problem 13 is a complete board with five piles.

- [2] **Problem 14.** Consider the chip-firing game played on a complete board with n piles. Consider the initial configuration where one pile has no chips, a second pile has one chip, a third pile has two chips, and so forth until the last pile is given $n - 1$ chips. (Since every pile is connected to every other, it is irrelevant which pile is which.) Prove that this is a “garden of eden” configuration. That is, the only way to arrive at this distribution of chips by firing a pile from another configuration is for that other configuration to also have one pile of zero chips, one pile of one chip, $\dots$, one pile of $n - 1$ chips.

- [3] **Problem 15.** Prove that the configuration in problem 14 is the *only* configuration of $0 + 1 + \dots + (n - 1) = \frac{n(n-1)}{2} = C$ chips on a complete board that never becomes stable (upto rearrangement of the piles, which, as pointed out in the previous problem, is irrelevant by symmetry).

Chip-firing games are a relatively new mathematical development, with development starting in the mid-1980's. Physicists study the *sandpile model*, which is essentially the same concept, as a way to learn about “self-organized criticality,” which is the ability of many physical systems to find their own stable states. There are many nice references on the Web to get a deeper look at this topic. A good print article is: A. Björner, L. Lovász and P. W. Shor, *Chip-firing game on graphs*, Europ. J. Combin., 12 (1991) 283-291.